

Designing Systems for Long-term Sensemaking

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Abstract

Personal informatics systems are widely used for self-tracking but often fall short in supporting long-term sensemaking, or how individuals derive meaningful insights from their data over extended periods of time. Existing tools overlook long-term routine lengths and disruptions and provide short-term feedback for users. This project addresses these limitations by investigating three key factors that inform the design of systems for long-term insights: routine lengths, routine disruptions, and the persistence of user-generated insights. Our investigation of longitudinal sleep records and systems that persist user insights provides key design considerations for supporting long-term sensemaking in personal informatics.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing design and evaluation methods**; *Empirical studies in HCI*.

Keywords

sensemaking; longitudinal study; long-term sensemaking; behavioral regularity; naturalistic study; personal informatics

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1 Introduction

Current personal informatics systems are not designed to support long-term sensemaking, or how individuals make sense of their data to gain insights over time [17, 28]. Existing systems are largely informed by short-term studies conducted in controlled, laboratory settings [6, 23, 26, 31, 32, 38, 39, 47, 52, 58] and are shaped by our limited knowledge of people’s long-term, naturalistic behaviors [6, 9, 31, 32, 38]. Such designs reinforce individuals to focus on short-term insights, making it challenging for researchers to understand how individuals make sense of their data in the long-term.

Thus, in current systems, there exist two major design challenges that prevent users from engaging in long-term sensemaking:

- (1) *Insufficient support for long-term exploration*. Existing personal informatics tools often overlook the structure and disruptions of long-term routines that shape behavior over time. Without tools that surface such long-term patterns, individuals are limited in their ability to derive long-term insights from their data.
- (2) *Reinforcement of short-term insights*. Existing systems are designed around short-term feedback cycles, or daily and weekly insights [7, 13–15, 18, 29, 42, 53], which may unintentionally reinforce short-term thinking [9]. As a result, by investigating existing systems, researchers lack a clear picture of how people reflect on and interpret their personal data over extended periods of time.

To address these interconnected challenges, we take a tripartite approach. To tackle the first hurdle, we investigate long-term naturalistic sleep records from 180,000 users from Sleep as Android. We examine sleep records as they serve as proxy signals for people’s overall daily patterns and provide insights into how individuals structure their broader routines [9]. In particular, we investigate two understudied aspects of routines: routine lengths [9] and routine disruptions. Next, we develop a system to help investigate the longitudinal user patterns and sensemaking practices people use to gain long-term insights from pre-existing self-tracking applications.

To validate whether such design considerations facilitate long-term sensemaking, we apply lessons from both investigations and develop a system incorporating routine lengths, routine disruptions, and sensemaking behaviors. The system enables participants to explore visualizations in three key ways: (1) customizing or filtering views based on routine lengths, (2) comparing behaviors before and after routine disruptions, and (3) demonstrating their sensemaking patterns over time. To evaluate whether the system helps users derive meaningful, long-term insights, participants will compare their experiences using our system with those from their existing tracking applications.

2 Related Work

Naturalistic behaviors reveal contextual factors and real-world constraints that shape people’s lives in ways we would not observe in controlled clinical settings [3, 16, 20, 33, 42, 51]. For example, routine disruptions demonstrate new sleep transitions for polyphasic sleep hackers [33] and biological changes for menstruators [20, 43, 51]. Such information enables systems to deliver timely interventions [4, 18, 44, 48] and help individuals gain a deeper understanding of themselves to develop actionable plans for behavioral change [9, 11, 12, 15, 19, 33–35].

Advancements in long-term sensemaking in personal informatics have been limited by the practical challenges of conducting long-term studies. Longitudinal investigations require significant



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labor investments to collect data [6, 31, 38, 39]. While ecological momentary assessments offset such costs [1], the method captures brief snapshots instead of a continuous picture of the individual. Long-term case studies offer deep insights into an individual's journey [16, 23, 53], but the method is limited in its scalability.

The widespread adoption of self-tracking technologies [2, 9, 10, 21, 22, 24, 25, 49, 50, 57] alleviates some of these costs and allows researchers to reveal behavioral patterns that unfold only through continuous, long-term observations. Researchers can now conduct large-scale investigations on routine phenotypes [36, 56] and investigate people's long-term behaviors in their naturalistic settings [9, 23, 26, 30, 39, 47, 52, 56, 58, 59]. As a result, we can address gaps in our understanding about long-term routines and disruptions, and their significance to individuals — knowledge to inform the design of systems for long-term sensemaking.

Furthermore, current personal informatics systems use calendar-based visualizations that overlook individual routine variations [9, 33, 34]. Such designs create inequitable experiences for users with 'non-normative' routine patterns as they prevent individuals whose natural rhythms do not align with these fixed timeframes from gaining meaningful insights [9, 12, 29, 34, 46, 55].

Users also lose valuable short-term insights, as they are designed to be ephemeral. For example, notifications containing information on user trends and highlights disappear once viewed and cannot be tagged or bookmarked for future analyses. Current commercial self-tracking applications (e.g. Apple Health, SleepCycle, Sleep as Android, Whoop, Oura) provide persistent visualizations of the data [2, 10, 21, 22, 24, 49, 50, 54, 57], but lack the mechanisms in which users can 'save' or 'favorite' the insights they find important.

Thus, how users engage with and understand insights, rather than raw tracking data, over time, particularly in the context of self-tracking, is understudied. Previous works focus on how to design data visualizations to encourage healthy behaviors [5, 6, 29] or support self-reflection on personal data [45]. Other works investigate how users externalize insights through data visualizations more broadly [8, 27, 40, 41].

Unlike prior works, our system treats insights as first-class data types, enabling users to save and manage them over time. This design allows us to observe the behavioral and cognitive patterns involved in sensemaking and identify key factors needed to create long-term sensemaking systems.

3 Methodology

This project aims to investigate three key aspects that can inform the design of systems to support long-term sensemaking. We incorporate the design considerations learned from studies on routine lengths, routine disruptions, and sensemaking routines.

3.1 Design Considerations

3.1.1 Routine Lengths. Our previous work demonstrated that visualization systems incorporating natural routine lengths enable users to discover meaningful insights and patterns that would otherwise remain hidden in their data [9]. We analyzed routine lengths of 180,000 Sleep as Android users by identifying each individual's longest streak of records and calculating the power spectral density to determine their most descriptive routine length. The findings

revealed a need to design for non-weekly routines, as the data suggested more than half the users followed non-weekly routines, including ones that extended over several months at a time [9].

3.1.2 Routine Disruptions. This work aims to demonstrate how routine disruptions in tracking data reflect meaningful, contextual factors and behavioral changes important for long-term insights, rather than random missingness. We hypothesize that these missing records and their lengths correlate with meaningful behavioral or contextual factors, such as elevated stress levels or major life events, and may indicate the emergence of new routines or the decay of a routine [3, 37, 48]. Absences in data could serve as meaningful signals to users and support long-term sensemaking.

To achieve this, we will first analyze participants' sleep tracking data from Sleep as Android to identify and characterize lapses or disruptions in their routines. For each identified disruption, we will compare behavioral patterns before and after the lapse to assess whether significant routine changes occur surrounding these gaps.

Next, to investigate associations between disruption characteristics, such as the duration of missing records, and major life changes, we will conduct a retrospective survey with the participants who exhibited substantial data gaps. The survey will ask participants to recall and explain the reasons for the missing tracking periods, thereby validating whether users retain memory and understanding of their own routine disruptions. By understanding the causes of missing data, or the absence of certain user behaviors, we can treat these gaps as meaningful signals that help us accommodate for diverse user scenarios, ultimately leading to the design of more robust real-world systems.

3.1.3 Routines in Sensemaking. This investigation underlines the value of designing systems that make personal insights persistent and accessible over the long-term. Saving such insights allows users to compute their own recording behaviors, such as the frequency of saved insights. Incorporating such features can help users identify which observations hold lasting significance.

Over a two-month period, participants will use our application to save ephemeral insights from their existing self-tracking applications through screenshots and annotations, allowing them to review a timeline of their insights over time. To understand the importance of saving insights, we will conduct a comparative study in which participants first use a prototype that does not display a timeline of their saved insights and one with a timeline visualization.

Preliminary pilot studies revealed two distinct types of sensemaking behaviors among users: one type of behavior oriented towards affirmation and validation [19], and the other driven by hypothesis formation and self-experimentation [12, 14]. We also find that users repeatedly mark the same observations as important. Although frequent recording indicates they have gathered enough evidence to establish knowledge about themselves, they often lack awareness on how frequently they assign significance to these insights. As a result, they continue tracking the same behaviors and miss opportunities to initiate action and change. Thus, saving insights supports long-term reasoning by helping users move beyond existing self-knowledge and take the next steps for their self-experimentation.

3.2 Designing a Long-term Sensemaking System

Based on our prior findings, we develop a web-based system designed to support long-term sensemaking from self-tracking data. The system allow users to import existing data from commercial self-tracking platforms. We will conduct a two-month study with 20 participants who have at least two years of tracking history to ensure a rich, longitudinal dataset.

The interface will enable users to filter and customize views based on their natural routine lengths and to compare behavioral patterns before and after routine disruptions. Additionally, the system will support persistent insight tracking and visualization. Users will also be able to save, annotate, and revisit their personal insights over time and keep a local copy of their data.

In addition, we will incorporate a novel sensemaking feature that leverages a large language model to process and visualize the logical reasoning trajectories behind users' recorded insights. This feature aims to surface the implicit causal connections and motivations that underlie a user's self-experimentation. The system will analyze the user's insights and annotations to generate a visual map, such as a Gantt Chart, to depict the user's temporal and conceptual progression of their reasoning over time. Such visualizations can help users identify the origins and development of their hypotheses and understand how their personal goals change over time. This feature will be tested in the latter half of the two-month study.

4 Evaluation

At the conclusion of the two-month study, we will conduct a mixed-methods evaluation to compare participants' experiences using our system with their previous experiences using commercial self-tracking tools. Through semi-structured interviews, we will qualitatively assess whether the system enabled users to generate new insights, uncover patterns they had not previously noticed, and initiate sustained behavioral changes. We will ask the user to compare the application with and without the sensemaking feature, and ask them to rank their existing commercial tracking application, the new system, and the new system with the sensemaking feature.

In parallel, we will analyze the content of saved insights and interaction logs across participants to identify patterns in how users engage in long-term sensemaking. This analysis will illuminate common behavioral and cognitive patterns and variations in how individuals use long-term sensemaking tools to derive value from their self-tracking data.

5 Expected Contributions

We expect the following contributions to lay the groundwork for more personalized and robust personal informatics systems:

- (1) Deepening understanding of how to design personal informatics systems that support long-term sensemaking. Our findings offer actionable design implications for commercial tracking tools that help users retain, revisit, and build upon meaningful insights over time.
- (2) Informing the design space of behavioral regularity and scheduling technologies. Our findings on routine lengths can inform how systems should align with users' real-world rhythms and coordinate across individuals. Our analysis of routine disruptions highlights the importance of treating

missing data as meaningful signals to consider for developers and designers of health technologies.

- (3) The deployed application enables users to import existing self-tracking data, visualize long-term routines and disruptions, and save persistent insights. These tools will allow others beyond the study to engage in long-term experimentation and self-discovery.

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